

Duality beyond rigidity

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Joint work with

Jürgen Fuchs, Gregor Schaumann, Christoph Schweigert, arXiv:[2306.17668](#), [2405.20811](#)

Robert Allen, Simon Lentner, Christoph Schweigert, arXiv:[2107.05718](#)

Tensor Categories, Quantum Symmetries, and Mathematical Physics
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Rigidity

Definition

- ① An object c in a monoidal category $(\mathcal{C}, \otimes)$ admits a *rigid right dual* $c^\vee$, if there exist morphisms

$$\mathrm{ev}^r : c^\vee \otimes c \rightarrow \mathbf{1}, \quad \mathrm{coev}^r : \mathbf{1} \rightarrow c \otimes c^\vee,$$

satisfying zig-zag relations.

- ② *Rigid left duals* are defined similarly.
- ③ An object $c \in \mathcal{C}$ is called *rigid* if it admits both a left and a right dual.

Rigid duals existing is a **property**, not structure.

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Examples

- 1 Category of finite-dimensional modules over a finite-dimensional Hopf algebra.
- 2 Category of admissible modules over a rational c_2 -cofinite vertex operator algebra.

Consequences of rigidity

- 1 In abelian rigid monoidal categories monoidal products are **exact**.
- 2 Projective objects form a monoidal ideal.
- 3 Taking duals is opmonoidal: $(c \otimes d)^\vee \cong d^\vee \otimes c^\vee$, that is $(-)^\vee : (\mathcal{C}, \otimes) \rightarrow (\mathcal{C}^{\text{opp}}, \otimes^{\text{opp}})$.

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Non-Examples

- 1 For A a finite-dimensional associative algebra and $B \in A\text{-Bimod}$. Then $B \otimes_A -$ is generally only **right exact**.
- 2 V a vertex operator algebra and $M \in V\text{-Mod}$ (to which HLZ theory applies). Then $M \otimes_V -$ is generally only **right exact**.

An extended example

A an associative unital algebra, $B, \tilde{B}$ bimodules. All finite-dimensional.

- The tensor product $B \otimes_A \tilde{B}$ is constructed by the coequaliser

$$B \otimes_{\mathbb{K}} A \otimes_{\mathbb{K}} \tilde{B} \rightrightarrows B \otimes \tilde{B} \rightarrow B \otimes_A \tilde{B} \rightarrow 0$$

and is hence **right exact**. Monoidal unit is A .

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- If B is an A -bimodule, then the linear dual $B^* = \text{Hom}_{\mathbb{K}}(B, \mathbb{K})$ is also an A -bimodule with action characterised by

$$(a_2 \cdot \beta \cdot a_1)(b) = \beta(a_1 \cdot b \cdot a_2), \quad \beta \in B^*, b \in B, a_i \in A.$$

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In general, $A^* \not\cong A$. Smallest example:

$$A_3 = \mathbb{K}[x, y] / \langle x^2, y^2, xy \rangle = \text{span}_{\mathbb{K}}\{[1], [x], [y]\}.$$

Admits a unique simple module S , with $[1] = \text{id}_S$, $[x] = 0$, $[y] = 0$.

$$0 \longrightarrow S \oplus S \longrightarrow A_3 \longrightarrow S \longrightarrow 0,$$

$$0 \longrightarrow S \longrightarrow A_3^* \longrightarrow S \oplus S \longrightarrow 0.$$

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A an associative unital algebra, $B, \tilde{B}$ bimodules. All finite-dimensional.

- A^* is a coalgebra and B is an A^* -bicomodule with coactions

$$\begin{aligned}\rho_r : B &\rightarrow B \otimes_{\mathbb{K}} A^*, \\ b &\mapsto b.(-)\end{aligned}$$

$$\begin{aligned}\rho_l : B &\rightarrow A^* \otimes_{\mathbb{K}} B, \\ b &\mapsto (-).b\end{aligned}$$

If $\{a_i\}$ is a basis of A with dual basis $\{\alpha_i\}$, then $b.(-) = \sum_i (b.a_i) \otimes_{\mathbb{K}} \alpha_i(-)$.

- The equaliser

$$0 \rightarrow B \otimes^A \tilde{B} \rightarrow B \otimes_{\mathbb{K}} \tilde{B} \rightrightarrows B \otimes_{\mathbb{K}} A^* \otimes_{\mathbb{K}} \tilde{B}$$

defines a **left exact** monoidal product (the cotensor product) with monoidal unit A^* .

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- We have

$$\mathrm{Hom}_A(B \otimes_A \tilde{B}, A^*) = \mathrm{Hom}_A(B, \tilde{B}^*).$$

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$$A_3 \otimes_{A_3} A_3 \cong A_3, \quad A_3 \otimes^{A_3} A_3 \cong S^{4\oplus}.$$

Grothendieck-Verdier categories / *-autonomous categories

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$$X \mapsto \mathrm{Hom}(X \otimes Y, K)$$

is representable. That is, there exists some $GY \in \mathcal{C}$ such that

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A *Grothendieck-Verdier* (GV) category is a monoidal category $(\mathcal{C}, \otimes)$ together with a choice of dualising object $K \in \mathcal{C}$.

Basic properties and examples

Proposition [Boyarchenko, Drinfeld '13]

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Theorem [Allen-Lentner-Schweigert-SW, '21]

Let V be a vertex operator algebra and $\mathcal{C}$ a category of V -modules (HLZ applies, $V \in \mathcal{C}$, closed under graded duals).

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Theorem [Allen '23], [Demirdelik '24]

Let H be a Hopf algebroid with finite-dimensional base algebra A and an invertible antipode S . Then the category of finite-dimensional H -modules is a GV category. A dualising object is given by the vector space dual of the base algebra A .

Important consequences

The dualising functor G is generally not opmonoidal for $\otimes$, but we can define a second monoidal product so that it is:

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The monoidal unit for $\boxtimes$ is K and $G : (\mathcal{C}, \otimes) \rightarrow (\mathcal{C}^{\text{opp}}, \boxtimes^{\text{opp}})$ is monoidal.

In the associative algebra example: $\boxtimes = \otimes^A$

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Now consider $\mathcal{C}$ as a monoidal category with **two** monoidal products: $\otimes$ and $\boxtimes$.

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Proposition

- ① $- \otimes Y$ admits a right adjoint (internal Hom):

$$\mathrm{Hom}(X \otimes Y, Z) \cong \mathrm{Hom}(X, \underline{\mathrm{Hom}}(Y, Z)), \text{ with } \underline{\mathrm{Hom}}(Y, Z) \cong Z \boxtimes GY.$$

- ② $Y \boxtimes -$ admits a left adjoint (internal coHom)

$$\mathrm{Hom}(X, Y \boxtimes Z) \cong \mathrm{Hom}(\underline{\mathrm{coHom}}(Y, X), Z), \text{ with } \underline{\mathrm{coHom}}(Y, X) \cong GY \otimes X.$$

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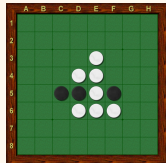
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Similar formulae for adjoints of $X \otimes -$, $- \boxtimes Z$.

Any statement about one tensor product has a dual version using the other. Like *Othello/Reversi* stones that can be flipped to change colour or a photo negative.



Module categories

Motivation

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- The construction of bulk/full/local CFTs requires module categories.
- Frobenius algebras constructed from module categories describe boundary field objects in CFT.
- Separate mathematical question: It has been asserted that GV categories admit distributors:

$$\delta^r : X \otimes (Y \boxtimes Z) \rightarrow (X \otimes Y) \boxtimes Z, \quad \delta^l : (X \boxtimes Y) \otimes Z \rightarrow X \boxtimes (Y \otimes Z)$$

These associate mixed tensor products and are generally **not** isomorphisms.

Module categories

All results from now on are from [Fuchs, Schaumann, Schweigert, SW '24].

Definition

A *left GV-module category* over a GV category $(\mathcal{C}, \otimes, K)$ is a left module category $(\mathcal{M}, \triangleright)$ such that the functors

$$- \triangleright m : \mathcal{C} \rightarrow \mathcal{M}, \text{ and } c \triangleright - : \mathcal{M} \rightarrow \mathcal{M}$$

both admit **right adjoints** (this property was called **closure** in Kenichi Shimizu's talk). Denote the respective right adjoints by $\underline{\text{Hom}}(m, -)$ and $H_c(-)$.

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Proposition

Let $(\mathcal{M}, \triangleright)$ be a left GV-module category over $(\mathcal{C}, \otimes, K)$, then the functor

$$\blacktriangleright : \mathcal{C} \times \mathcal{M} \rightarrow \mathcal{M}, \quad c \blacktriangleright m = H_{Gc}(m)$$

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Remark

Just like the two tensor products $\otimes, \boxtimes$, the two actions are mutually adjoint.

$$\mathrm{Hom}(m, c \blacktriangleright n) \cong \mathrm{Hom}(Gc \triangleright m, n)$$

The unit and counit of these adjunctions define **coevaluation** and **evaluation**, respectively.

$$\mathrm{ev}_{c,m} : Gc \triangleright (c \blacktriangleright m) \rightarrow m \quad \mathrm{coev}_{c,m} : m \rightarrow c \blacktriangleright (Gc \triangleright m)$$

Module functors

Let $\mathcal{M}, \mathcal{N}$ be GV module categories over $(\mathcal{C}, K)$ and let $F : \mathcal{M} \rightarrow \mathcal{N}$ be a functor.

Def: Module structures

- A *lax* $\triangleright$ -module structure on F is a natural transformation

$$f_{c,m}^{\triangleright} : c \triangleright F(m) \rightarrow F(c \triangleright m)$$

that is compatible with the $\triangleright$ -module constraints of $\mathcal{M}$ and $\mathcal{N}$. Same for lax $\blacktriangleright$ -module structures.

- An *oplax* $\triangleright$ -module structure on F is a natural transformation

$$f_{c,m}^{\triangleright} : F(c \triangleright m) \rightarrow c \triangleright F(m)$$

that is compatible with the $\triangleright$ -module constraints of $\mathcal{M}$ and $\mathcal{N}$. Same for oplax $\blacktriangleright$ -module structures.

- A *strong* $\triangleright$ -module structure on F is a natural isomorphism

$$f_{c,m}^{\triangleright} : c \triangleright F(m) \cong F(c \triangleright m)$$

that is both lax and oplax.

Transporting module structures

Let $\mathcal{M}, \mathcal{N}$ be GV $(\mathcal{C}, K)$ -module categories and $F : \mathcal{M} \rightarrow \mathcal{N}$ a functor.

Proposition

The following module structures are in bijection.

- Conjugation: lax $\triangleright$ on $F \leftrightarrow$ oplax $\blacktriangleright$ on F .

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- Flipping: lax $\triangleright$ on $F \leftrightarrow$ lax $\blacktriangleright$ on G
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- Profunctor: $\text{oplax } \triangleright \text{ on } F \leftrightarrow \text{lax } \triangleright \text{ on } G$
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- Flipping: $\text{lax } \triangleright \text{ on } F \leftrightarrow \text{lax } \blacktriangleright \text{ on } G$ Uses the adjunction.
 $\text{oplax } \triangleright \text{ on } F \leftrightarrow \text{oplax } \blacktriangleright \text{ on } G$
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- Profunctor: $\text{oplax } \triangleright \text{ on } F \leftrightarrow \text{lax } \triangleright \text{ on } G$ Uses profunctor transport of [Shimizu '23].
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Appropriate compositions of flipping and profunctor transport give conjugation.

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Definition/example

The functors which apply module objects from the right

$$R_m^{\triangleright} : \mathcal{C} \rightarrow \mathcal{M},$$

$$c \mapsto c \triangleright m$$

$$R_m^{\blacktriangleright} : \mathcal{C} \rightarrow \mathcal{M}$$

$$c \mapsto c \blacktriangleright m$$

are strong $\triangleright, \blacktriangleright$ -module functors with module structure given by the associativities of $\triangleright, \blacktriangleright$. The conjugates of these module structures are oplax $\blacktriangleright$ / lax $\triangleright$ -module structures called **distributors**.

$$\delta_{c,d,m}^r : (c \otimes d) \triangleright m \rightarrow c \blacktriangleright (d \triangleright m), \quad \delta_{c,d,m}^l : c \triangleright (d \blacktriangleright m) \rightarrow (c \otimes d) \blacktriangleright m$$

Distributor pentagons

Proposition

Unlike associators, distributors need not be isomorphisms. They satisfy 6 associator-like pentagons.

$$\begin{aligned}\delta^r \text{ only: } & ((c \otimes d) \otimes e) \triangleright m \Rightarrow c \blacktriangleright (d \blacktriangleright (e \triangleright m)) \\ & (c \otimes d) \triangleright (e \triangleright m) \Rightarrow c \blacktriangleright (d \triangleright (e \triangleright m))\end{aligned}$$

$$\begin{aligned}\delta^l \text{ only: } & (c \otimes d) \triangleright (e \blacktriangleright m) \Rightarrow (c \otimes (d \otimes e)) \blacktriangleright m \\ & c \triangleright ((d \otimes e) \blacktriangleright m) \Rightarrow (c \otimes d) \blacktriangleright (e \blacktriangleright m)\end{aligned}$$

$$\begin{aligned}\text{mixed: } & (c \otimes (d \otimes e)) \triangleright m \Rightarrow (c \otimes d) \blacktriangleright (e \triangleright m) \\ & (c \otimes d) \triangleright (e \blacktriangleright m) \Rightarrow (c \otimes (d \otimes e)) \blacktriangleright m\end{aligned}$$

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$$\begin{aligned}\delta^l \text{ only: } & (c \otimes d) \triangleright (e \blacktriangleright m) \Rightarrow (c \otimes (d \otimes e)) \blacktriangleright m \\ & c \triangleright ((d \otimes e) \blacktriangleright m) \Rightarrow (c \otimes d) \blacktriangleright (e \blacktriangleright m)\end{aligned}$$

$$\begin{aligned}\text{mixed: } & (c \otimes (d \otimes e)) \triangleright m \Rightarrow (c \otimes d) \blacktriangleright (e \triangleright m) \\ & (c \otimes d) \triangleright (e \blacktriangleright m) \Rightarrow (c \otimes (d \otimes e)) \blacktriangleright m\end{aligned}$$

Previously observed for GV categories in [Cockett, Seely '97] through direct computation.

Distributor pentagons

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The approach via module functors adds conceptual clarity, simplifies the proof and generalises the result. Here the proof of the first 4 of these pentagons follows immediately from the weak module pentagons of the distributors. The proof of the mixed pentagons requires more care but is still short.

(Co)evaluation and zig-zag relations

Proposition

Consider $(\mathcal{C}, K)$ as a GV-module (so $\triangleright = \otimes$, $\blacktriangleright = \boxtimes$). Define

$$\mathrm{ev}_c : Gc \otimes c \cong Gc \otimes (c \boxtimes K) \xrightarrow{\mathrm{ev}_{c,K}} K, \quad \mathrm{coev}_c : 1 \xrightarrow{\mathrm{coev}_{c,1}} c \boxtimes (Gc \otimes 1) \cong c \boxtimes Gc.$$

Then $\mathrm{ev}_c, \mathrm{coev}_c$ can also be characterised as

$$\mathrm{Hom}(1, c \boxtimes Gc) \cong \mathrm{Hom}(Gc, Gc) \cong \mathrm{Hom}(Gc \otimes c, K).$$

$$\mathrm{coev}_c \mapsto \mathrm{id}_{Gc} \mapsto \mathrm{ev}_c$$

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These GV (co)evaluations satisfy **zig-zag relations**.

$$\begin{array}{ccccccc} c & \xrightarrow{\mathrm{coev}_c \otimes \mathrm{id}} & (c \boxtimes Gc) \otimes c & \xrightarrow{\delta^r} & c \boxtimes (Gc \otimes c) & \xrightarrow{\mathrm{id} \boxtimes \mathrm{ev}_c} & c \\ & \searrow \mathrm{id}_c & & & & & \nearrow \end{array}$$

$$\begin{array}{ccccccc} Gc & \xrightarrow{\mathrm{id} \otimes \mathrm{coev}_c} & Gc \otimes (c \boxtimes Gc) & \xrightarrow{\delta^l} & (Gc \otimes c) \boxtimes Gc & \xrightarrow{\mathrm{ev}_c \boxtimes \mathrm{id}} & Gc \\ & \searrow \mathrm{id}_{Gc} & & & & & \nearrow \end{array}$$

Distinguished subcategories

The functors $\underline{\text{Hom}}(m, -) = (- \triangleright m)^{r.a.}$ and $\underline{\text{coHom}}(m, -) = (- \blacktriangleright m)^{l.a.}$ are naturally **weak** module functors by profunctor transport.

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Definition

- $m \in \mathcal{M}$ is called $\otimes$ -admissible if
 - ① $\underline{\text{Hom}}(m, -)$ is a **strong** module functor over $(\mathcal{C}, \otimes)$.
 - ② $\underline{\text{Hom}}(m, -)$ admits a right adjoint.
- $\widehat{\mathcal{M}}^\otimes$ is the full subcategory of $\otimes$ -admissible objects.
- $\otimes$ -admissibility is defined analogously using $\underline{\text{coHom}}(m, -)$
- Considering $\mathcal{C}$ as a module category over itself yields $\widehat{\mathcal{C}}^\otimes, \widehat{\mathcal{C}}^\otimes$.

The categories $\widehat{\mathcal{M}}^\otimes, \widehat{\mathcal{M}}^\otimes$ are generally not abelian even if $\mathcal{M}$ is. Note that $1 \in \widehat{\mathcal{C}}^\otimes$ and $K \in \widehat{\mathcal{C}}^\otimes$.

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Proposition

- 1 The subcategories $\widehat{\mathcal{C}}^\otimes, \widehat{\mathcal{C}}^{\boxtimes}$ are unital monoidal subcategories in $(\mathcal{C}, \otimes)$ and $(\mathcal{C}, \boxtimes)$.
- 2 $\widehat{\mathcal{M}}^\otimes$ and $\widehat{\mathcal{M}}^{\boxtimes}$ are module categories over $\widehat{\mathcal{C}}^\otimes, \widehat{\mathcal{C}}^{\boxtimes}$.

Algebras and coalgebras

For any $m \in \mathcal{M}$,

- Internal End $A_m = \underline{\text{Hom}}(m, m)$ is an **associative unital algebra** in $(\mathcal{C}, \otimes)$.
- Internal coEnd $C_m = \underline{\text{coHom}}(m, m)$ is a **coassociative counital coalgebra** in $(\mathcal{C}, \otimes)$.
- Natural module functor

$$\begin{aligned}\underline{\text{Hom}}(m, -) : \mathcal{M} &\rightarrow \text{Mod-}A_m \\ n &\mapsto \underline{\text{Hom}}(m, n)\end{aligned}$$

Same for $\underline{\text{coHom}}(m, -)$ and comodules.

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Proposition

If $\mathcal{C}, \mathcal{M}$ are finite abelian, then

- 1 For $m \in \mathcal{M}$, the functor $\underline{\text{Hom}}(m, -)$ is a module equivalence if and only if $m \in \widehat{\mathcal{M}}^{\otimes}$ and m is a $\mathcal{C}$ generator of $\mathcal{M}$.
- 2 Similar for $\underline{\text{coHom}}(m, -)$.

Frobenius algebras

Proposition

Let $(A, \mu, \eta) \in (\mathcal{C}, \otimes)$ $\mu : A \otimes A \rightarrow A, \quad \eta : \mathbf{1} \rightarrow A,$

be a unital associative algebra. Then the following are in bijection.

- 1 Coalgebra structures $\Delta : A \rightarrow A \otimes A, \epsilon : A \rightarrow K$ such that

$$(\mu \otimes \text{id}_A) \circ \delta_{A,A,A}^l \circ (\text{id}_A \otimes \Delta) = \Delta \circ \mu = (\text{id}_A \otimes \mu) \circ \delta_{A,A,A}^r \circ (\Delta \otimes \text{id}_A).$$

- 2 Non-degenerate morphisms $\kappa : A \otimes A \rightarrow K$ such that

$$\kappa \circ (\mu \otimes \text{id}_A) = \kappa \circ (\text{id}_A \otimes \mu) \circ \alpha_{A,A,A}.$$

- 3 Frobenius forms $\lambda : A \rightarrow K$ (related to isomorphisms $A \rightarrow GA$).

Serre functors

We wish to consider objects $m \in \mathcal{M}$ such that their internal homs or cohomomorphisms are representable as internal cohomomorphisms or homs (this is called internal representability). That is,

$$\underline{\mathrm{Hom}}(m, -) \cong \underline{\mathrm{coHom}}(Sm, -), \quad \text{or} \quad \underline{\mathrm{coHom}}(m, -) \cong \underline{\mathrm{Hom}}(\tilde{S}m, -).$$

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Proposition

- 1 $\underline{\mathrm{Hom}}(m, -)$ is internally representable if and only if m is $\otimes$ -admissible ($m \in \widehat{\mathcal{M}}^{\otimes}$).
Then $Sm = \underline{\mathrm{Hom}}(m, -)^{r.a.}(K)$.
- 2 $\underline{\mathrm{coHom}}(m, -)$ is internally representable if and only if m is $\boxtimes$ -admissible ($m \in \widehat{\mathcal{M}}^{\boxtimes}$).
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Then $\tilde{S}m = \underline{\mathrm{coHom}}(m, -)^{l.a.}(1)$.
- 3 S is the **relative Serre functor** of $\mathcal{M}$ and $\tilde{S}$ is the **inverse relative Serre functor** of $\mathcal{M}$.
These functors are mutually inverse:

$$\widehat{\mathcal{M}}^{\otimes} \begin{array}{c} \xrightarrow{S} \\ \xleftarrow{\tilde{S}} \end{array} \widehat{\mathcal{M}}^{\boxtimes}$$

Serre functors and Frobenius algebras

Proposition

- 1 The relative Serre functor of $\mathcal{C}$ is canonically a monoidal equivalence $S_{\mathcal{C}} : \widehat{\mathcal{C}}^{\otimes} \rightarrow \widehat{\mathcal{C}}^{\boxtimes}$ and
$$S_{\mathcal{C}}(-) = G^2(-) \otimes K, \quad \widetilde{S}_{\mathcal{C}}(-) = G^2(-) \boxtimes 1.$$
- 2 The relative Serre functor $S_{\mathcal{M}}$ of $\mathcal{M}$ is a twisted module functor
$$S_{\mathcal{M}}(c \triangleright m) \cong S_{\mathcal{C}}(c) \blacktriangleright S_{\mathcal{M}}(m), \quad c \in \widehat{\mathcal{C}}^{\otimes}, m \in \widehat{\mathcal{M}}^{\otimes}.$$
- 3 Let $m \in \widehat{\mathcal{M}}^{\otimes}$. For every isomorphism $p : m \rightarrow Sm$ there is a natural Frobenius algebra structure on $\underline{\text{Hom}}(m, m)$. Note that the existence of such a morphism p implies $m \in \widehat{\mathcal{M}}^{\boxtimes}$.

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The above is a generalisation of [Shimizu '23] where Serre functors for exact module categories over pivotal finite tensor categories were considered.

Example for oldies/experienced people (or why I am excited about all this)

In [Gaberdiel-Runkel-SW '09] the $c = 0$ triplet, $W_{2,3}$, and its category of modules $W_{2,3}\text{-mod}$ were considered.

$$0 \rightarrow W(2) \rightarrow W_{2,3} \rightarrow W(0) \rightarrow 0$$

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Conjecturally, that category consists of those objects $c \in \widehat{W_{2,3}\text{-mod}}^{\otimes}$ for which an isomorphism $c \cong Sc$ exists.

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1	$(0), 1, 7$	$(0), 2, 5$	$\frac{1}{3}, \frac{10}{3}$
2	$\frac{5}{8}, \frac{33}{8}$	$\frac{1}{8}, \frac{21}{8}$	$\frac{-1}{24}, \frac{35}{24}$

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- Perhaps these Serre functor fixed points are the correct starting point for boundary CFT.

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- Grothendieck-Verdier duality is a natural duality structure for many algebraic representation categories, including vertex algebras.
- There is a natural theory of module categories over Grothendieck-Verdier categories that provides many of the features needed for full/bulk/local conformal field theory.
- Next: Try and construct conformal field theories from Grothendieck-Verdier categories and their modules.

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Thank you for coming!